

A Sustainable House for the Canadian Arctic

Many housing models that have been introduced in the north do not adequately reflect the culture and lifestyle of northern peoples or communities.

by **Bill Semple**

With dramatic rises in the cost of energy, it has become clear to northern Canadian governments, housing agencies, homeowners, and renters that the way we build and operate homes is unsustainable. Furthermore, climate change may have a dramatic impact on Inuit culture in the Canadian Arctic. We need to design and construct buildings that significantly reduce energy use, which contributes to global warming, and at the same time are adaptable to the climate change already taking place in the Far North.

In addition, there is a growing recognition that many housing models that have been introduced in the north do not adequately reflect the culture and lifestyle of northern peoples or communities. For example, the Inuit traditionally did not have a home as we know it; they often migrated and made use of snow houses in winter and skin houses in summer. But the homes in which the Inuit live today are modern homes, where they live year-round.

To address these issues, Canada Mortgage and Housing Corporation (CMHC), where I work, partnered with northern housing providers to carry out the design, construction, monitoring, and evaluation of two northern housing prototypes. The goal of the project is to design a culturally appropriate Northern Sustainable House that consumes 50% less energy than the Model National Energy Code

for Houses (MNECH). The project incorporated a design process that included charrettes (problem-solving workshops) with members of the local communities, builders, housing corporation representatives, and technical experts.

Two models for the Northern Sustainable House have been initiated by this project—a model for Inuit communities in the eastern and northern Arctic developed in partnership with the Nunavut Housing Corporation (NHC) in Arviat, Nunavut, and a model in the western Arctic developed in partnership with the Tr'ondek Hwech'en First Nations Community in Dawson City, Yukon. This article focuses on the first of these models. The Nunavut project gave CMHC an opportunity to build upon other work involving partnerships between the NHC and CMHC. This includes the design work carried out by the NHC on the development of a five-unit housing complex designed to perform 25% better than the MNECH.

The Arviat House Design Process

Joe Karatek, an Inuit facilitator from the Department of Education in Arviat, worked with me to facilitate the Northern Sustainable House design charrette, which was held on September 27–29, 2006, although our original plan was to meet for one day only. Charrettes



The Greenland House, Arviat, Nunavut, includes a cold storage room near the entrance for storing animal skin clothing.

are often used to create forums for the free flow of ideas, to bring issues to the table and encourage thinking outside of the box. As charrettes address a variety of needs (planning, design, technical, social/cultural, and so on), the goal of generating ideas can be at odds with the desire to quickly solve the technical and other problems posed by these ideas. Not surprisingly, this occurred the first morning of the Arviat House design charrette.

To facilitate discussion, while making sure that technical problems would also be addressed, we decided to modify the charrette process by scheduling a second session for the following day to address technical issues. This ensured that issues with the implementation of the design ideas would be addressed and allowed all of the participants to get into the idea-generating process of the charrette. Later on the first day, as it became apparent that the perspectives of women were not being fully defined or addressed, an additional half-day session was organized for the third day. This final session brought together several Inuit women from the community.

To carry out the design of the Northern Sustainable House, we organized a CMHC/NHC design team, led by myself. The design team began its work by identifying several ideals and principles that needed to be addressed. While the house must be adaptable to local cultural needs and present climatic conditions, it was also

weatherization



Here are some crated windows at the job site. Materials are shipped in once a year by sea lift.

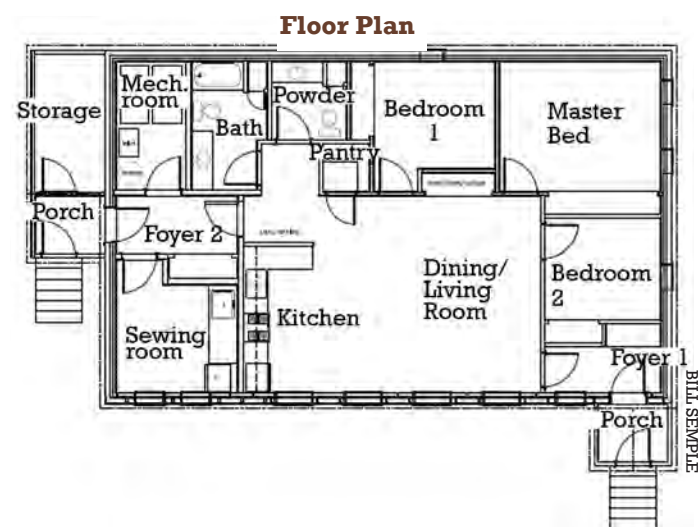


Figure 1. The team began by evaluating and consolidating the cultural design ideas that were identified in the charrette and incorporating these features into the design of the Sustainable Northern House.

important that the house be easily and inexpensively modified if community needs change and as the climate changes. This meant that the design must be based on simple construction details that would make the home easily adaptable and as cost-effective as possible.

In addition to climate change mitigation features, such as improved energy efficiency, which reduces carbon emissions and the use and high cost of oil shipped to the Far North, the design needed to allow for the easy incorporation of alternative energy technologies, either at the time of construction or to be added later.

began by evaluating and consolidating the cultural design ideas that were identified in the charrette and incorporating these features into the design of the Sustainable Northern House. The most important of these features are described below (see Figure 1).

A large, open living room/dining room/kitchen. The Inuit divide their eating into two different kinds of foods that are eaten in the house—country foods (raw) and southern foods (cooked). When Inuit gather together to eat country foods,

Since the costs of many technologies will be reduced as economies of scale in their development kick in, the house had to be built to be ready to make use of active solar-thermal and/or PV technology.

Initial Design

The initial work of the design team focused on developing the floor plan and design features of the house. The team

the meals are eaten on the floor, while cooked foods are eaten at the table. This is partly because country foods are set out in piles of raw meat, which each person carves and cuts up as he or she eats. Many houses are designed with small kitchens for southern families. These kitchens do not provide enough space for larger extended families to sit together and eat country foods.

Two entrances. We designed the Inuit house to have a summer and a winter entrance, at the urging of the Inuit elders, who have lived through many hard winters. In recent homes built for Inuit families, doors are often placed without considering the prevailing winds. As a result, cold drafts and direct winds enter the house, and high snow drifts form at the entrance. One participant in the charrette commented that in spite of fire regulations, he put a tarp over his north vestibule in the winter each year to try to keep the cold winds out. Even when they received more insulation and new siding, these houses often remained cold. In the winter, the entrance is best located in the west. In the summer the entrance is best located in the south.

Warm, cool, and cold spaces.

Some animal skins, such as polar bear and fox, cannot be worked on indoors, because these skins have a very strong smell and the fur comes off easily in the warm house; these animals are skinned outside on a plastic ground sheet with a sheet of cardboard laid over it. When Inuit carry out skinning and carving inside the house, they do it on the floor. The Inuit women in the charrette highlighted some key points with respect to dressing skins. Locating the laundry tub near the entrance of the house, and near the place where the skins are to be sewn, would help to confine the mess to a small area. If the temperature in houses is too warm and the humidity is too low, the hair on most skins, if they are kept in a warm room for too long, will begin to fall out. At the same time, the skins cannot be frozen, or they will become too difficult to sew. The optimum temperature for keeping skins

weatherization

is about 2°C– 5°C (36°F–41°F). For sewing skins, the women needed a cool room that could be maintained at this temperature.

In addition to the cool room, the Inuit need a cold storage area for storing skin clothing. In Arviat, a model called the Greenland House provides a good example of such a room. The cold storage room in the Greenland House is located adjacent to the entrance to the house.

An isolated mechanical room.

By isolating the mechanical equipment from the rest of the house, the design team attempted to solve two problems. First, many Inuit complain about the noise of mechanical rooms. Second, the housing authorities need easy access to the mechanical equipment in order to do maintenance and repairs, and putting the equipment in one room makes this easy and less disruptive to the home occupants.

A laundry room or an area with a large laundry tub. A large tub is needed for soaking skins. Locating the laundry tub near the entrance—ideally in a separate laundry room—would help to confine the mess, as explained above.

In addition, several other features were brought into the design. Some of these features were identified in other charrettes or in conversations with northerners about housing. Others were introduced as ways to make the house more flexible and adaptable to the occupants' future needs:

- There should be a large kitchen pantry area for food storage, as a great deal of bulk food from the south arrives during the summer shipping season only.
- There should be a separate desk/counter space for use as a working area adjacent to the living area. This would provide a space for students to work at their computers and so on, without being isolated from the rest of the family. Elders have often spoken of the importance of having



A computer-generated image shows an interior perspective (view towards south west corner) of the Northern Sustainable House.

young people near, so that they will hear the stories of the elders and learn more about the traditions of their people.

- Transom windows should be installed over bedroom doors to bring more daylight into the rooms from the bright south-facing living/kitchen area.
- Space should be provided for an additional half bathroom or additional storage (depending on the needs of the family).
- Exterior door framing should be built into the wall at a suitable access point, at the time of construction. This makes it easier to add another bedroom to the house in the future, if one is needed.

The cost of construction is a major concern in the Far North; both materials and labor cost 2 to 3 times as much as they cost in south Canada. Other features were brought into the design to address this issue:

- Hallways and corridors were reduced or eliminated, and the common space of the house was correspondingly enlarged.
- The kitchen, bathrooms, and laundry areas were clustered to reduce the length of piping runs. This reduces construction costs by reducing the materials needed. It also reduces operating costs, because the shorter supply runs reduce hot water losses.

- The structural layout was simplified to reduce building costs and to ensure that construction could be easily carried out by local builders, with the standard building materials used in the community.

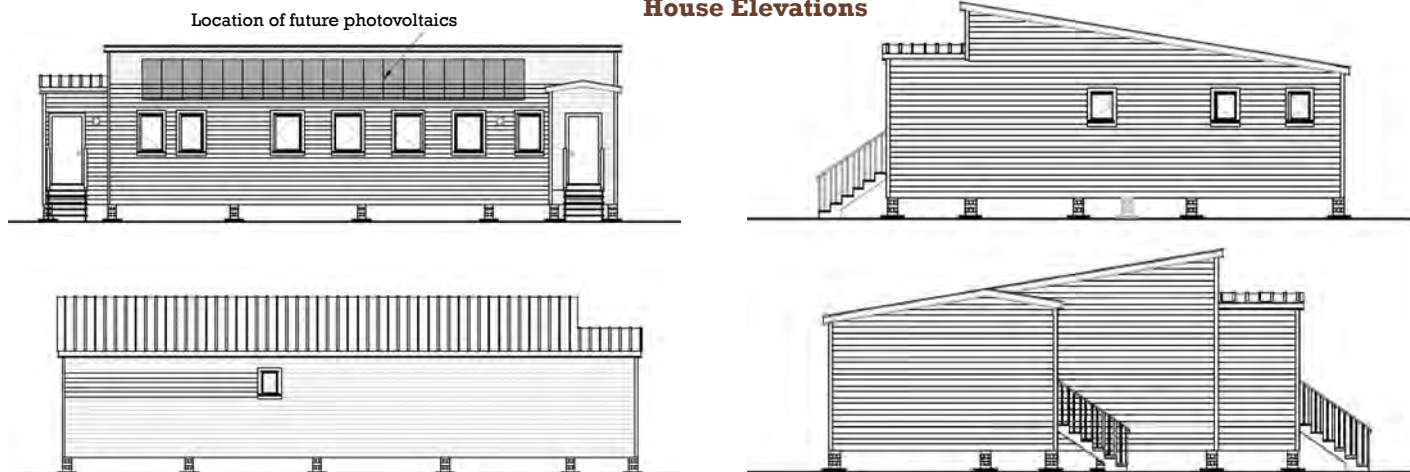
Seeking More Input

A second workshop, to evaluate, and get local input on, the design was held in Arviat on November 30, 2006 with the original participants in the September 2006 charrette. This was done to ensure that what was being developed reflected the ideas that had been raised in the charrette. The participants responded very positively to the design, and they suggested several ways to improve upon it.

Participants acknowledged that the house very effectively addressed the needs of the Inuit family, inside a modest footprint. The elders noted that northern living is always based on understanding the wind and the sun, and they said that the design did this very well. The Inuit also appreciated the fact that they had been asked to contribute to the design process; they said that this had not happened before. Donald Uluadluak commented that the whole experience had left him with the feeling of being in a newly built igloo, and “the smell of new snow and sense of accomplishment that comes with building it.”

In moving forward we all also felt that the ventilation system must be very well designed, and especially that it

House Elevations



Figures 2 through 5. The South Elevation (top left), North Elevation (bottom left), East Elevation (top right), and West Elevation (bottom right) of the Northern Sustainable House.

must take into account the number of people in the house.

The suggestions put forth at this workshop were incorporated into the design. On December 5, 2006, the revised design was presented to Peter Scott, president of the Nunavut Housing Corporation, and others at the head office of the NHC in Iqaluit (see Figures 2–5).

Energy-Efficient Design Features

As many reports from the Far North have stated, perhaps nowhere has felt the impact of climate change more strongly than the polar regions of the planet. Observations on climate change and the work of many researchers and practitioners agree that climate change will require builders to adopt both mitigation and adaptation strategies—with housing being no exception. While the features described up to this point were developed to address cultural issues, the designers of the Arviat House put equal effort into addressing issues of energy efficiency. They also wanted to provide a design that would use—or could easily be adapted to use—active solar-thermal or PV technology.

The designers were also motivated by the need to reduce the high costs of importing heating oil from southern Canada, to reduce the operating costs of the house. As explained above,

the project established the goal of developing a house that consumes 50% less energy than the MNECH. This work built upon previous work carried out by the NHC and CMHC on the development of the NHC 5-Plex, which was designed to perform 25% better than the MNECH. To reach the 50% better than the MNECH goal, the design team explored alternative wall and ceiling construction systems, and other innovations that would be needed to meet the energy conservation targets for the project. As part of this process, we carried out energy modeling on each of the systems to evaluate the expected energy savings to be realized by each.

Based on current building practices, input from the NHC and CMHC, and ideas raised in the design charrette, several wall systems were designed, and two of these were developed as options for the project. Wall system A consists of structural insulated panels (SIPs). Additional strapping, a vapor barrier, and insulation would be added at the building site after assembly to ensure long-term airtightness and improved energy performance. The SIPs option was evaluated for its potential to speed up the process of construction, particularly in smaller isolated communities.

The second wall system (wall system B) consisted of a double-framed wall constructed using conventional framing techniques and skills used by

the Nunavut Housing Corporation (see Figure 6). Both systems were developed, evaluated, and recommended for the project. In both cases, the floor insulation was maintained at R-40, with ceiling insulation increased from R-40 to R-67. The existing wall insulation level of approximately R-28 was increased to approximately R-50 in wall system A and R-48 in wall system B.

To further reduce energy consumption, the following other features were introduced into the design:

- The house was given a southern orientation, and the size of the windows was increased on the south wall and reduced on all other elevations, particularly on the north wall.
- The design specified triple-glazed, argon-filled, fiberglass windows.
- The elevation was reduced to 7 feet on the north side of the house to reduce northern exposure, and a shed roof was used to deflect the prevailing northwest winter winds.
- The higher elevation on the south side of the house was used to provide space for the installation of PV solar water-heating panels above the south windows, and to allow increased daylight penetration into the house (see image, opposite page). The house is designed to be solar ready. (It is important to note

Table 1. Analysis of Arviat House, Nunavut

Option	Consum MJ/yr	R2000 Consum MJ/yr	R2000 Target MJ/yr	Elec KWH /yr	Elec Cost \$/yr	Oil Litres L/yr	Oil Cost \$/yr	Energy MJ Save/yr	Energy % of Base	Energy Cost \$/yr	Energy Save \$/yr
MNECH—78% Oil	299,599	268,026	90,683	17,367	\$9,378	6,154	\$6,154	0	100%	\$15,532	\$0
MNECH—71% Oil	339,322	307,749	90,683	17,405	\$9,398	7,172	\$7,172	-39,723	113%	\$16,570	-\$1,038
Existing 2.0 ach ₅₀	183,213	151,666	90,683	16,647	\$8,990	3,200	\$3,200	116,386	61%	\$12,190	\$3,342
Existing 1.5 ach ₅₀	175,025	143,478	90,683	16,610	\$8,969	2,991	\$2,991	124,574	58%	\$11,960	\$3,572
Type A	111,833	88,172	90,683	10,527	\$5,685	1,919	\$1,919	187,766	37%	\$7,604	\$7,928
Type B	114,893	91,231	90,683	10,539	\$5,691	1,998	\$1,998	184,706	38%	\$7,689	\$7,843
Column 1	Lights + Appliances + Heat + Hot Water										
Column 2	Heat + Hot Water Only (R2000 Target Criteria)										
Column 3	R2000 Target (Heat + Hot Water), equivalent to EnerGuide 80										

here that vertical installations work well in the Far North, due to the lower sun angles. This is especially true when there is an additional potential for solar gain from sun sunlight reflected off the snow during the extended winter.)

- High-energy efficient florescent lighting fixtures were specified throughout the house.
- The bathroom, laundry, and kitchen areas were clustered to reduce hot water consumption, as described above.
- A high-energy-efficient oil boiler was used to provide heat and hot water, and a high-efficiency heat recovery ventilator (HRV) was used for heat recovery.
- The house was equipped with a low-water-consumption stacking washer/dryer.
- The design specified an isolated foyer for each entrance to reduce heat loss.
- The design specified advanced framing details to reduce construction materials and thermal bridging.

Energy Modeling

To better understand the potential energy performance of the house at the design stage, a consultant was hired to work with the design team to carry out energy modeling; to calculate the energy savings to be realized by the different combinations of wall and ceiling systems; and to ensure, at the design stage, that the project would achieve its stated goal of consuming

50% less energy than the MNECH. This included modeling the wall system used at present by the NHC to provide a base from which to calculate expected energy savings.

The results of the modeling are shown in Table 1. These results show that both wall system A and wall system B attained the targeted energy performance levels for the project. Wall system A attained a reduction of 63% (consuming 37% of total heating energy). Wall system B attained a reduction of 62% (consuming 38% of total heating energy). Anticipated energy costs in either case would be reduced by approximately \$4,500 per year over present practices and approximately \$7,800 over the MNECH.

In addition to improving the energy performance of the house, the double-wall system developed for this project is potentially stronger and more durable than the present wall system used for the construction of houses by the NHC. It is anticipated that, as the impact of climate change becomes more dramatic, the additional strength and durability of the house could increase both longevity and performance.

While these results are difficult to qualify, we hope that the use of the integrated design process, and the design of housing that is more culturally appropriate for the occupants and the community, will foster a greater sense of ownership of the building among the Inuit, and increase occupants' willingness to maintain and repair their housing. This is a significant issue in a region where a very high percentage of

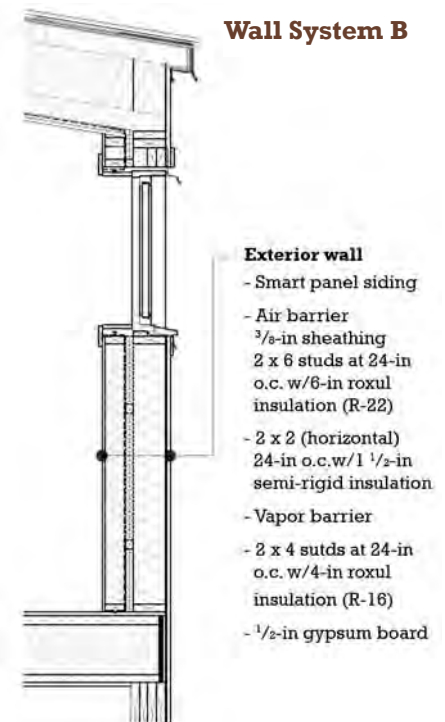


Figure 6. The second wall system (wall system B) consisted of a double-framed wall constructed using conventional framing techniques and skills.

housing is provided or subsidized by the Territorial Government.

Construction and Performance Monitoring

From the final design plans, a materials list is being prepared for the construction of two models of the Northern Sustainable House—one using each of the wall systems designed and evaluated for the project. The materials will be shipped to arrive in the summer of 2008, and construction is scheduled to take place during the latter part of the 2008 building season. When construction is complete, and

occupants have moved into the house, the performance of the house will be monitored for a period of 12 months. This will include gathering data on the amount of energy used for space heating and domestic hot water, electricity consumption, and water consumption, as well as data on humidity levels and indoor air quality.

The same monitoring protocol will be carried out on an NHC house in Arviat built using current construction practices. A comparison of the cost of construction of the two houses and the actual energy saved will be carried out to determine the monthly carrying costs of the additional features of the Sustainable Northern House versus the NHC Conventional House. The high cost of construction in the north, the tremendous housing shortage, and limited capital funds are consistently cited by northern housing agencies as significant impediments to incorporating additional energy features into the construction of northern housing. One of the objectives of the project is to demonstrate that the annual savings in operating costs can cover these additional capital costs. As different departments or divisions within departments handle capital and operating costs within separate budgets, knowing the numbers will help everyone realize that these additional features can be paid for.

In addition to evaluating the performance of the house, CMHC will survey occupants of the house, participants in the design charrette, staff from the NHC, and other people in the community to gauge their reaction to the Northern Sustainable House and to the design process used to develop it. This information will provide important feedback on the best way to carry out ongoing work in this area. (Look for a follow-up article in *Home Energy*.)

Where Do We Go from Here?

Like others involved in this project, I see it not as something final, but more as a work in progress. More study, more design charrettes, and more evaluations are needed to further develop the cultural features of the house. The effort here is to create something that can be successful, to show that it works in a variety of ways, and to build on that. Too many initiatives have been introduced from the outside and have failed. I have learned that the process is as important as the outcome, and that, as outsiders to this region, we must first learn to respect what is there.

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